

Describing Meaning Exercises: Lattices, Plurals, and Mass Terms

Conventions.

- $\leq$ is an order relation. In linguistic applications, $\leq$ will usually be interpreted as *part-of*.
- $\sqcap$ = meet (greatest lower bound), $\sqcup$ = join (least upper bound).
- $+$ is used for sums of individuals (often coinciding with join under part-of).

Exercises

Exercise 1

Let S be the set of **non-empty** subsets of $\{1, 2, 3\}$, ordered by $\subseteq$.

1. List all members of S .
2. Draw the Hasse diagram for $(S, \subseteq)$.
3. For each pair below, compute the join $A \sqcup B$ (if it exists in S):

$$(\{1\}, \{2\}), (\{1\}, \{1, 2\}), (\{2, 3\}, \{3\}).$$

4. For each pair below, compute the meet $A \sqcap B$ (if it exists in S):

$$(\{1\}, \{2\}), (\{1, 2\}, \{2, 3\}), (\{1, 3\}, \{3\}).$$

5. Give one explicit example of a pair (A, B) for which the meet fails to exist in S , and explain why.
6. What single element could you add to S so that every pair has a meet?
7. After adding that element, is the resulting structure a lattice? Explain.

Exercise 2

Let $S = \{1, 2, 3, 4, 6, 12\}$ ordered by divisibility ($x \leq y$ iff $x \mid y$).

1. Draw the Hasse diagram.
2. Compute $4 \sqcap 6$ and $4 \sqcup 6$ (within S).

3. Compute $3 \sqcap 4$ and $3 \sqcup 4$ (within S).
4. Compute $2 \sqcap 12$ and $2 \sqcup 12$ (within S).
5. Which elements are the minimal elements of this poset? Which element is maximal?
6. Give a one-sentence description of what meet and join correspond to here.

Exercise 3

Fix a world w with exactly four atomic **hippos**:

$$\llbracket \text{hippo} \rrbracket^w = \{A, B, C, D\}.$$

Assume plural sums exist for any combination, and interpret $x \leq y$ as x is a part of y .

1. Draw the Hasse diagram for the structure containing the four atoms and all their non-atomic sums.
2. List all *immediate* supersums of A (i.e. the elements that cover A in your diagram).
3. List all parts of $A + B + C + D$.
4. Compute $(A + B) \sqcap (B + C)$ and $(A + B) \sqcup (B + C)$.
5. Compute $(A + C) \sqcap (C + D)$ and say whether the meet is an atom or a non-atom (or neither).

Exercise 4

Continue with the same world w and atoms A, B, C, D .

1. Give a denotation (as a set) for $\llbracket \text{hippos} \rrbracket^w$, assuming it denotes the non-atomic hippo sums.
2. Give a denotation for $\llbracket \text{exactly two hippos} \rrbracket^w$.
3. Give a denotation for $\llbracket \text{less than three hippos} \rrbracket^w$.
4. Assume definites pick out the maximal sum of relevant atoms in the situation. What should $\llbracket \text{the hippos} \rrbracket^w$ be?

Exercise 5

Let G be a portion of **mud**. Suppose G has two proper parts G_1 and G_2 such that:

$$G_1 \leq G, \quad G_2 \leq G, \quad \text{and} \quad G = G_1 + G_2.$$

Assume downward closure:

$$\text{mud}(x) \ \& \ y \leq x \Rightarrow \text{mud}(y).$$

1. If $\text{mud}(G)$, what follows about $\text{mud}(G_1)$ and $\text{mud}(G_2)$?

2. Suppose also that G_1 itself has proper parts H_1 and H_2 with $G_1 = H_1 + H_2$. What follows about $mud(H_1)$ and $mud(H_2)$?
3. Explain why downward closure conflicts with the idea that there is a linguistically privileged “one mud” atom.

Exercise 6

Consider:

1. *more gravel*
2. *more marbles*

1. What do these normally compare (volume/weight vs cardinality vs something else)?
2. Give a context where *more marbles* could be judged by weight rather than count.

Exercise 7

Let *statue* be a count predicate (statues as atoms and plural sums), and let *clay* be a mass predicate (downward closed). Assume a mapping *mat* from statue-individuals to clay-portions that is sum-preserving:

$$mat(x + y) = mat(x) + mat(y).$$

1. Describe a scenario where two distinct statues $S_1 \neq S_2$ have the same material portion: $mat(S_1) = mat(S_2)$.
2. Would sum-preservation rule that out? Why not?
3. Propose one additional constraint on *mat* that would make it harder (or impossible) for distinct statues to map to exactly the same clay portion.

Exercise 8

Let $Atom_P(x)$ mean: $P(x)$ and x has no proper P -parts.

1. Give a natural definition of $Atom_P(x)$ using $\leq$ (part-of) and P .
2. Which of the following plausibly have atoms in the semantic sense: *marble*, *furniture*, *rice*, *committee*, *sand*?
3. Pick one hard case and give two competing analyses (one atomic, one non-atomic), each with a linguistic diagnostic.